Oladipupo Ridwan Bello, PhD

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SUMMARY

Computational immunologist specializing in building predictive and generative machine learning models of the immune system from single-cell multi-omics and structural biological data. Focused on accelerating the discovery of broad-spectrum and targeted immunotherapies for human and animal health.

EDUCATION

PhD in Animal Sciences (Genetics and Genomics), University of Maryland, College Park, USA, 2025
Dissertation: Exploring the Drivers of Public T-Cell Receptors Using Deep Learning and Transcriptomics
Credential Verification: https://registrar.umd.edu/ceValidate.html (CeDiD: 267W-D70Z-ONOW)

MSc in Animal Breeding & Genetics (Distinction), University of Ibadan, Nigeria, 2019

BSc in Animal Science (First Class Honors), University of Ibadan, Nigeria, 2015

RELEVANT SKILLS

- Programming: R, Python, Bash, SQL, Go, C, Rust, TypeScript.
- Bioinformatics: Bulk/single-cell RNA-Seq, WGS, ChIP-Seq, ATAC-Seq, TCR/BCR repertoire analysis, structural modeling, metagenome analysis, GWAS, genomic selection.
- Statistical Analysis: GLMs & mixed models, survival analysis, multivariate statistics, Bayesian inference.
- Machine Learning: Random forests, deep learning, generative models, classification, clustering.
- Laboratory Techniques: Microscopy, DNA extraction, PCR, gel electrophoresis, molecular cloning.
- Environments and Tools: Git, Docker, Singularity, HPC/Slurm, Nextflow, Quarto.

RESEARCH EXPERIENCE

Department Lead, Healthcare Department (Volunteer) (January 2026 - Present)
Community Dreams Foundation (cdreamstream.org)

- Direct the software development and scientific validation of three health-technology platforms focused on general health and athlete performance.

Doctoral Research Assistant (January 2021 - December 2025)
Department of Animal and Avian Sciences, University of Maryland, College Park, USA

- *Identified molecular and structural drivers of public (shared) T cell receptors in human and bovine immunity*
  - Built a convolutional neural network classifier achieving over 85% precision and recall to distinguish public from private T cell receptor nucleotide sequences.
  - Modeled 3-dimensional T-cell receptor structures in MHC contexts, revealing structural determinants associated with receptor sharing.
  - Profiled public T cell transcriptomes from single-cell RNA-Seq data, characterizing their distinct cytotoxic effector phenotypes.
  - Mapped intercellular communication networks and analyzed the relationship between gut microbial and T cell receptor diversity.
  - Identified candidate genes and linked public T cell receptor signatures to complex traits through functional enrichment and GWAS summary statistics integration.

- *Identified genetic and epigenetic mechanisms of resistance to Marek's disease virus-induced T cell lymphoma in chickens*
  - Identified and validated candidate genes associated with lymphoma resistance by integrating bulk

  RNA-Seq, ChIP-Seq, and ATAC-Seq chromatin accessibility profiles.
  - Detected RNA editing events implicated in lymphoma resistance by exploiting the highly inbred nature of the chicken lines.
  - Characterized biological functions and pathways of co-expressed candidate genes through co-expression network analysis.

- Served as Laboratory Compliance Officer, ensuring the lab met university safety and regulatory standards.

TEACHING EXPERIENCE

Graduate Teaching Assistant (January 2021 - December 2025)
Department of Animal and Avian Sciences, University of Maryland, College Park, USA

- *ANSC 101/103 (Principles of Animal Science & Lab)*: Demonstrated animal handling and veterinary care.
- *ANSC 627/327 (Molecular & Quantitative Genetics)*: Conducted regular office hours and review sessions.
- *ANSC 447 (Physiology of Mammalian Reproduction Lab)*: Led labs on gross anatomy and histology.

Tutorial Assistant (March 2017 - September 2018)
Department of Animal Science, University of Ibadan, Nigeria

- Led tutorials on molecular/quantitative genetics and introductory statistics for undergraduate students.

PUBLICATIONS

Bello, O.R., Arvind, J., & Johnson, P.L.F. (In preparation). Intrinsic Determinants of TCR Publicness and Their Structural Implications. *PLOS Computational Biology*.

Bello, O.R. & Johnson, P.L.F. (In preparation). Distinct Cytotoxic Effector Phenotypes of Public T Cells. *Frontiers in Immunology*.

Bello, O.R., Salako, A.E., Akinade, A.S., & Yakub, M. (2023). Relationship between Milk Yield and Udder Morphology Traits in White Fulani Cows. *Dairy*, 4, 435-444. [Journal]

Ewuola, E.O., Adeyemi, A.A., & **Bello, O.R.** (2020). Variations in haematological and serum biochemical indices among White Fulani bulls, Ouda rams and West African Dwarf bucks. *Nigerian Journal of Animal Production*, 44(1), 136-143. [Journal]

Bello, O.R., & Johnson, P.L.F. (January 10-15, 2025). *Public T Cell Receptors in Bovine Immunity*. International Plant & Animal Genome Conference, San Diego, California, USA. [Abstract]

Bello, O.R., Chu, Q., & Song, J. (July 10-13, 2023). *Temporal profiling of the bursa transcriptome reveals systemic differences induced by Marek's disease virus*. PSA, Philadelphia, Pennsylvania, USA. [Abstract]

SELECTED HONORS AND AWARDS

- Invited Peer Reviewer (2025): Elsevier
- Jacob K. Goldhaber Travel Grant (2025): Graduate School, University of Maryland, College Park, USA
- Animal Health & Care Academy Fellow (2024/2025): MANRRS, USA (sponsored by Merck & Zoetis)
- Shaffner Award for Second Place in Poultry Research (2022): 35th Annual Symposium, Department of Animal and Avian Sciences, University of Maryland, College Park, USA
- Dean's Fellowship (2021): Graduate School, University of Maryland, College Park, USA
- EducationUSA Opportunity Funds Program Fellow (2019): United States Embassy in Nigeria
- University of Ibadan Master's Scholarship & Tutorial Assistantship (2017): University of Ibadan, Nigeria
- Overall Best (Nationwide), Graduate Animal Scientist Exam (2016): Nigerian Institute of Animal Science
- Best Student, Agriculture Practical Year Training Program (2014): University of Ibadan, Nigeria